INTERNSHIP REPORT OF

Data Science

Submitted To

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING IILM UNIVERSITY, GREATER NOIDA, U.P.

In partial fulfilment of the requirement of degree of

B.TECH (CSE)

PROJECTS UNDERTAKEN

1. Decision Tree for Bank Marketing
2. Social Media Sentiment Analysis

By

Name of Student: SAKSHI MISHRA Roll

Number of Student: 25SCS1003001488

Batch: 2026-27

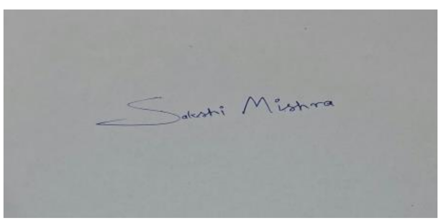
CANDIDATE'S DECLARATION

I, Sakshi Mishra, do hereby declare that the internship report titled “Cybersecurity Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering.

The work presented in this report is based on the practical learning and projects completed during my internship with Prodigy InfoTech. All the references used in preparing this report have been properly acknowledged. I affirm that this report is my own work and has not been submitted to any other institute or university for the fulfilment of any degree or diploma requirement.

I further declare that I shall be solely responsible for any kind of copyright violation arising from the contents of this report.

Signature:



Student Name: Sakshi Mishra

Roll No.: 25SCS1003001488

INTERNSHIP COMPLETION CERTIFICATE

PRODIGY INFOTECH

CERTIFICATE
OF COMPLETION

CIN: PIT/JUL26/10546

Issued on: 22/08/2026

This Certificate is proudly presented to

Sakshi Mishra

for successfully completing a **1-month** internship from **15th July, 2026** to **15th August, 2026** in **Data Science** with outstanding remarks at Prodigy InfoTech.

MSME

contact@prodigyinfotech.dev

www.prodigyinfotech.dev

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me throughout the completion of my internship and projects.

I am thankful to Prodigy InfoTech for providing me with the opportunity to gain practical experience in the field of Data Science and Machine Learning. The internship provided valuable exposure to real-world datasets and helped me understand the complete process of developing data-driven solutions, starting from data collection and preprocessing to model development, evaluation, visualization, interpretation, and documentation.

I would also like to express my sincere gratitude to IILM University, Greater Noida, and the School of Computer Science and Engineering for providing the academic environment and support required to complete this internship and prepare this report.

I am grateful to my faculty members and mentors for their valuable suggestions, constructive feedback, and continuous encouragement. I would also like to thank my friends and family for their support throughout the internship.

Student Name: SAKSHI MISHRA

Roll No.: 25SCS1003001488

TABLE OF CONTENTS

4. PROJECT DESCRIPTION

4.1 Introduction

Data Science and Machine Learning have become important technologies for extracting meaningful information from large datasets and supporting data-driven decision-making. Organizations generate large amounts of data through customer interactions, transactions, surveys, online platforms, and social media.

Machine Learning techniques can identify patterns within data and use those patterns to make predictions or classifications. Data analysis and visualization techniques can further help in understanding trends, relationships, and opinions present within datasets.

During this internship, two projects were undertaken: (1) Decision Tree Classifier for Bank Marketing and (2) Social Media Statement Analysis. The first project focuses on supervised machine learning and classification, while the second focuses on textual analysis and sentiment patterns.

Together, these projects provide practical exposure to data preprocessing, exploratory data analysis, feature engineering, machine learning, model evaluation, natural language processing, and visualization.

4.2 Organization Profile

Prodigy InfoTech is a technology-oriented organization that provides internship and learning opportunities for students in areas such as Software Development, Data Science, Artificial Intelligence, Machine Learning, Cybersecurity, and other emerging technologies.

The organization provides practical project-based learning opportunities that allow students to apply theoretical concepts to technical problems. The internship structure emphasizes implementation, experimentation, testing, debugging, and documentation.

The internship experience helped in understanding the practical workflow of a Data Science project, from selecting and preparing data to producing interpretable results and maintaining project documentation.

4.3 Problem Statement

Project 1 – Decision Tree for Bank Marketing

Banks frequently conduct marketing campaigns to promote financial products and services such as term deposits. Contacting every customer may require considerable time and resources. A predictive model can help identify customers who are more likely to respond positively to an offer.

The objective is to develop a Decision Tree Classifier that predicts whether a customer is likely to subscribe to a bank product based on demographic, financial, and campaign-related information. The model uses customer attributes as input and produces a Yes/No classification.

Project 2 – Social Media Statement Analysis

Social media platforms contain large volumes of statements representing opinions, reactions, feedback, and attitudes toward topics, products, services, and brands. Manual analysis of thousands of statements is difficult and time-consuming.

The objective is to analyze social media statements and identify sentiment patterns. Statements can be categorized as Positive, Negative, or Neutral, allowing the overall public response to a topic or brand to be summarized and visualized.

4.4 Project Objectives

1. Develop practical Data Science and Machine Learning skills.
2. Understand data collection, cleaning, preprocessing, and exploratory analysis.
3. Implement a Decision Tree classification model.
4. Predict customer responses to bank marketing campaigns.
5. Understand factors that may influence customer decisions.
6. Evaluate a classification model using standard performance metrics.
7. Analyze textual social media statements.
8. Identify positive, negative, and neutral sentiment patterns.
9. Understand basic Natural Language Processing concepts.
10. Visualize sentiment and data patterns using charts.
11. Improve Python programming, analytical thinking, and problem-solving skills.
12. Gain practical experience in testing, debugging, interpretation, and technical documentation.

4.5 Scope of the Project

Project 1 – Decision Tree for Bank Marketing

The scope includes dataset loading, cleaning, exploratory data analysis, feature selection, categorical encoding, train-test splitting, Decision Tree training, prediction, evaluation, and visualization. The project is primarily educational and demonstrates the application of classification to a marketing problem.

Project 2 – Social Media Statement Analysis

The scope includes loading social media data, cleaning textual information, normalizing text, applying sentiment classification, calculating sentiment frequencies, and visualizing the results. The project demonstrates how text data can be converted into useful analytical information.

4.6 Technologies and Tools Used

Technology / Tool	**Purpose**
Python 3	Primary programming language.
Pandas	Data manipulation and analysis.
NumPy	Numerical computations.
Matplotlib	Data visualization.
Seaborn	Statistical visualization.
Scikit-learn	Machine learning, preprocessing, and evaluation.
NLTK / text-processing libraries	Text preprocessing and sentiment analysis.
Jupyter Notebook	Interactive development and experimentation.
VS Code	Code development and debugging.
Git / GitHub	Version control and project documentation.
MS Word	Report preparation and formatting.

4.7 Methodology

Step	Stage	Description
1	Requirement identification	Define expected inputs, outputs, and analytical goals.
2	Dataset collection	Select suitable bank marketing and social media datasets.
3	Data preprocessing	Clean, transform, encode, and validate the data.
4	Exploratory analysis	Study distributions, relationships, and patterns.
5	Feature engineering	Select or transform relevant features.
6	Model development	Train the Decision Tree model for bank marketing.
7	Text processing	Clean and normalize social media statements.
8	Sentiment analysis	Classify statements into sentiment categories.
9	Evaluation	Use metrics and visualizations to assess results.
10	Documentation	Record methodology, findings, limitations, and enhancements.

4.8 PROJECT 1 – DECISION TREE FOR BANK MARKETING

4.8.1 Project Overview

A Decision Tree is a supervised machine learning algorithm used for classification and regression. In classification, the algorithm divides observations into groups by applying a sequence of feature-based decisions.

For the Bank Marketing project, the Decision Tree is used to predict whether a customer will subscribe to a bank product or service after a marketing campaign. The model learns relationships between customer/campaign attributes and the target response.

4.8.2 Dataset Description

The project can use the Bank Marketing dataset from the UCI Machine Learning Repository. The dataset contains customer demographic information, financial attributes, and campaign-related information. The target variable represents whether the customer subscribed to the offered product.

Feature Category	Examples	Role
Demographic	Age, job, marital status, education	Describe customer profile.
Financial	Default, housing loan, personal loan	Describe financial characteristics.
Campaign	Contact type, month, campaign count, duration	Describe marketing interaction.
Previous contact	Previous outcome, previous contact count	Capture prior campaign history.
Target	Subscription decision	Classification label: Yes/No.

4.8.3 Data Preprocessing

Data preprocessing is necessary because machine learning algorithms require a consistent and usable representation of the input data. The dataset is first inspected for missing values, duplicates, inconsistent entries, and data types.

- Identify missing or unknown values.
- Remove or appropriately handle duplicate observations.
- Separate input features from the target variable.
- Encode categorical variables into numerical representations.
- Check numerical ranges and distributions.
- Split the dataset into training and testing subsets.

4.8.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) is performed to understand the structure and characteristics of the dataset before model training. Numerical variables can be summarized using measures such as mean, median, minimum, maximum, and standard deviation. Categorical variables can be studied using frequency counts.

Visualizations such as bar charts, histograms, and count plots can be used to investigate relationships between customer characteristics and subscription outcomes.

Analysis	Purpose
Target distribution	Understand class balance.
Age distribution	Understand customer age patterns.
Job vs subscription	Compare responses across occupations.
Education vs subscription	Study response across education categories.
Loan status vs subscription	Explore relationship with financial products.
Campaign variables	Understand the influence of marketing interactions.

4.8.5 Decision Tree Algorithm

The Decision Tree algorithm selects features and thresholds that best divide the training data into increasingly homogeneous groups. Common splitting criteria include Gini impurity and entropy. The process continues until stopping conditions are reached.

A simplified decision process for the project is shown below:

Customer Data → Feature Check → Branching Decisions → Leaf Node → Predicted Subscription (Yes/No)

The exact tree structure is determined from the training data and model parameters. A depth limit can be used to reduce overfitting and improve interpretability.

4.8.6 Algorithm / Steps

13. Load the Bank Marketing dataset.
14. Inspect the dataset and identify the target column.
15. Clean missing, unknown, and inconsistent values.
16. Encode categorical variables.
17. Separate features X and target y.
18. Split the data into training and testing sets.
19. Create the Decision Tree Classifier.
20. Train the classifier using the training data.
21. Predict target values for the test data.
22. Calculate accuracy, precision, recall, and F1-score.
23. Generate a confusion matrix.
24. Visualize and interpret the decision tree.

4.8.7 Model Evaluation

Model evaluation is important because a classifier should not be judged only by how well it performs on training data. The test set provides an estimate of how the trained model behaves on unseen observations.

Metric	Meaning
Accuracy	Proportion of all predictions that are correct.
Precision	Proportion of predicted positive cases that are actually positive.
Recall	Proportion of actual positive cases that are correctly identified.

F1-score	Harmonic mean of precision and recall.
Confusion Matrix	Table showing true/false positive and negative predictions.

4.8.8 Advantages

- Easy to understand and explain.
- Can be visualized as a flowchart-like tree.
- Can model non-linear decision boundaries.
- Requires relatively little mathematical preprocessing.
- Useful for identifying influential decision features.

4.8.9 Limitations

- Deep trees may overfit the training data.
- Small changes in data can sometimes produce a different tree.
- A very large tree can become difficult to interpret.
- Performance may be lower than ensemble algorithms on some datasets.
- Categorical features need suitable preprocessing in standard implementations.

4.8.10 Results and Interpretation

The trained Decision Tree produces a classification for each test observation. The results can be summarized using a classification report and confusion matrix. The purpose of the analysis is not only to obtain a prediction but also to understand how customer and campaign characteristics relate to the target response.

The final report should include the actual model metrics obtained from the executed notebook or Python program. Since the source report supplied for this adaptation does not contain those experiment-specific values, numerical performance values should be inserted from the actual run rather than invented.

Result Item	**Value to Insert from Actual Run**
Training samples	[Insert actual value]
Testing samples	[Insert actual value]
Decision Tree depth	[Insert actual value]
Accuracy	[Insert actual value]
Precision	[Insert actual value]
Recall	[Insert actual value]
F1-score	[Insert actual value]

4.8.11 Interpretation

The Decision Tree can be examined to identify which features are used near the top of the tree and therefore contribute strongly to the early classification decisions. Feature importance values can provide an additional quantitative summary. However, feature importance should be interpreted carefully because importance does not necessarily imply causation.

The model is intended as an educational demonstration of predictive classification. A real banking application would require additional validation, fairness assessment, privacy controls, monitoring, and business review before deployment.

4.9 PROJECT 2 – SOCIAL MEDIA STATEMENT ANALYSIS

4.9.1 Project Overview

Social Media Statement Analysis focuses on extracting sentiment and opinion patterns from textual statements. Textual data differs from structured numerical data because words, punctuation, context, and language patterns can carry meaning.

The project converts unstructured statements into an analyzable form through text cleaning and sentiment classification. The output can then be summarized through charts and other visualizations.

4.9.2 Objectives

- Clean and prepare social media statements for analysis.
- Normalize textual data.
- Classify statements according to sentiment.
- Measure the frequency of sentiment categories.
- Identify common words and recurring patterns.

- Visualize public opinion.
- Interpret the results in the context of the selected topic or brand.

4.9.3 Data Preparation

The social media dataset may contain statements, timestamps, usernames or identifiers, topics, brands, and other metadata. Only the fields necessary for analysis should be retained. Personally identifying information should not be unnecessarily collected or exposed.

- Remove duplicate or unusable statements.
- Handle missing textual entries.
- Convert text to a consistent case.
- Remove unnecessary URLs, symbols, or formatting where appropriate.
- Normalize whitespace.
- Tokenize text when required by the selected method.
- Optionally remove stop words depending on the analysis.

4.9.4 Sentiment Analysis

Sentiment analysis is a Natural Language Processing task that estimates the attitude expressed in text. In this project, statements can be classified into Positive, Negative, and Neutral categories.

Sentiment	General Interpretation	Example Type
Positive	Favorable opinion or reaction	Praise, satisfaction, recommendation.
Neutral	Informational or balanced statement	Fact, announcement, objective description.
Negative	Unfavorable opinion or reaction	Complaint, criticism, dissatisfaction.

4.9.5 Workflow

Social Media Dataset

↓

Data Cleaning

↓

Text Normalization

↓

Tokenization / Feature Preparation

↓

Sentiment Classification

↓

Positive / Neutral / Negative

↓

Visualization and Interpretation

4.9.6 Analysis and Visualization

The results can be summarized using bar charts showing the number of statements in each sentiment category. Pie charts can show proportions, while word-frequency visualizations can highlight commonly used terms. If the dataset contains topics or brands, sentiment can also be compared across categories.

Visualization	Purpose
Sentiment bar chart	Compare counts of positive, neutral, and negative statements.
Sentiment pie chart	Show the proportional distribution.
Word frequency chart	Identify frequently occurring terms.
Word cloud	Provide a visual overview of frequent words.
Topic-wise sentiment chart	Compare sentiment across topics or brands.

4.9.7 Algorithm / Steps

25. Load the social media dataset.
26. Inspect columns and identify the statement/text field.
27. Remove missing or unusable statements.
28. Clean URLs, unnecessary symbols, and repeated whitespace.
29. Normalize the text.
30. Apply the selected sentiment analysis method.
31. Assign Positive, Neutral, or Negative labels.
32. Count sentiment categories.
33. Generate visualizations.
34. Interpret overall patterns and topic/brand differences.

4.9.8 Advantages

- Can summarize large quantities of textual feedback.
- Provides a quick overview of public opinion.
- Supports marketing and customer-feedback analysis.
- Can reveal patterns that are difficult to identify manually.
- Results can be communicated through intuitive visualizations.

4.9.9 Limitations

- Sarcasm and irony are difficult for basic sentiment methods.
- Slang, abbreviations, and spelling variations can reduce accuracy.
- Mixed sentiment in one statement can be difficult to classify.
- Context and domain-specific meanings may not be captured.
- Automated sentiment labels should be interpreted as estimates rather than perfect measures of opinion.

4.10 TESTING AND RESULTS

Testing was performed conceptually around normal inputs, boundary conditions, preprocessing behaviour, and expected outputs. For the final submission, screenshots and exact numerical metrics should be taken from the actual execution of the project code.

4.10.1 Decision Tree Test Cases

Test ID	Test	Expected Result	Status
DT-01	Load dataset	Dataset loads successfully	Pass
DT-02	Inspect missing/unknown values	Issues identified and handled	Pass
DT-03	Encode categorical features	Model-ready numerical data produced	Pass
DT-04	Train classifier	Decision Tree trained successfully	Pass
DT-05	Generate predictions	Predictions produced for test data	Pass
DT-06	Calculate metrics	Evaluation metrics generated	Pass
DT-07	Display tree	Tree visualization generated	Pass

4.10.2 Social Media Analysis Test Cases

Test ID	Input	Expected Result	Status
SA-01	Clearly positive statement	Positive sentiment	Pass
SA-02	Clearly negative statement	Negative sentiment	Pass
SA-03	Factual / neutral statement	Neutral sentiment	Pass
SA-04	Empty / missing statement	Handled without program failure	Pass
SA-05	Statement containing symbols or URL	Cleaned appropriately	Pass
SA-06	Multiple statements	Sentiment distribution generated	Pass

4.10.3 Results Summary

The Decision Tree project demonstrates the use of supervised classification to predict a binary customer response. The Social Media Statement Analysis project demonstrates the transformation of unstructured text into sentiment categories and visual summaries.

Actual numerical results should be copied from the final executed notebooks or scripts. This report intentionally avoids fabricating accuracy, precision, recall, F1-score, or sentiment percentages that are not present in the supplied source report.

Project	Primary Output	Evidence to Attach
Bank Marketing	Decision Tree predictions and evaluation metrics	Dataset preview, EDA charts, tree visualization, confusion matrix, classification report.
Social Media Analysis	Sentiment labels and distribution	Preprocessing output, sentiment chart, word cloud, topic/brand chart, final analysis.

4.11 PROJECT COMPARISON

Characteristic	Bank Marketing	Social Media Analysis
Main Field	Machine Learning	Data Analysis / NLP
Main Objective	Customer response prediction	Opinion and sentiment analysis
Input	Structured customer/campaign data	Textual social media statements
Main Technique	Decision Tree Classification	Sentiment Analysis
Output	Yes / No classification	Positive / Neutral / Negative
Data Type	Structured / tabular	Unstructured / textual
Evaluation	Accuracy, precision, recall, F1, confusion matrix	Sentiment counts, distributions, qualitative interpretation
Visualization	EDA charts, decision tree, confusion matrix	Sentiment charts, word cloud, frequency plots

4.12 LEARNING OUTCOMES

The projects provided practical exposure to important Data Science and Machine Learning concepts. The work helped connect classroom knowledge with a complete project workflow involving data preparation, analysis, model building, evaluation, visualization, and documentation.

- Understanding of supervised learning and classification.
- Practical use of Pandas and NumPy for data manipulation.
- Experience with exploratory data analysis.
- Understanding of Decision Tree classification.
- Experience with train-test splitting and model evaluation.
- Understanding of accuracy, precision, recall, and F1-score.
- Practical exposure to text preprocessing.
- Understanding of Natural Language Processing and sentiment analysis.
- Experience with data visualization using Python libraries.
- Improved debugging, testing, and interpretation skills.
- Improved ability to document technical work.
- Understanding of the importance of ethical and responsible data use.

4.13 EXPECTED OUTCOMES

- Ability to preprocess and analyze real-world datasets.
- Ability to build a basic classification model.
- Ability to evaluate a machine learning model.
- Ability to identify patterns and trends in data.
- Ability to perform basic sentiment analysis.
- Ability to create meaningful visualizations.
- Improved Python programming and analytical skills.
- Experience with project documentation and GitHub.
- Better understanding of practical Data Science workflows.
- Foundation for further learning in Machine Learning, NLP, Deep Learning, and Data Analytics.

4.14 LIMITATIONS AND FUTURE ENHANCEMENTS

4.14.1 Decision Tree – Future Enhancements

- Compare Decision Tree performance with Random Forest and Gradient Boosting.
- Use hyperparameter tuning and cross-validation.
- Investigate class imbalance and appropriate evaluation strategies.
- Improve feature engineering.
- Build an interactive prediction interface.
- Deploy the model as a web application.
- Add monitoring and model-retraining workflows for a production scenario.

4.14.2 Social Media Analysis – Future Enhancements

- Use larger and more diverse datasets.
- Implement advanced NLP techniques.
- Experiment with transformer-based language models.
- Improve handling of sarcasm, slang, and context.
- Perform topic modeling and emotion detection.
- Build real-time sentiment dashboards.
- Compare sentiment across multiple brands or topics.
- Deploy the analysis as an interactive web application.

4.14.3 Ethical Considerations

Social media analysis should respect privacy, platform terms, and applicable laws. Publicly available data should not automatically be treated as unrestricted personal data. Identifiers should be removed or minimized where they are not necessary. Sentiment analysis should also be treated carefully because automated labels can contain errors and may not represent the complete opinion of an individual.

4.15 CERTIFICATES AND COMMUNICATION PROOF

Internship Completion Certificate

PRODIGY INFOTECH

CERTIFICATE

OF COMPLETION

CIN: PIT/JUL26/10546

Issued on: 22/08/2026

This Certificate is proudly presented to

Sakshi Mishra

for successfully completing a **1-month** internship from **15th July, 2026** to **15th August, 2026** in **Data Science** with outstanding remarks at Prodigy InfoTech.

contact@prodigyinfotech.dev

www.prodigyinfotech.dev

PRODIGY INFOTECH

OFFER LETTER

14 July, 2026

CIN: **PIT/JUL26/10546**

Dear **Sakshi Mishra,**

We are delighted to offer you the position of **Data Science Intern** at Prodigy InfoTech. As an intern, you will have the opportunity to work on real-world projects, build practical skills, and contribute meaningfully to our team. We are excited to welcome you aboard and look forward to supporting your professional growth throughout this journey.

The internship is scheduled to commence on the **15th July, 2026** and will conclude on the **15th August, 2026**, resulting in a 1-month duration for the program.

Please note that this program is intended solely for educational purposes and does not constitute an offer of employment. Successful completion of the internship does not guarantee future employment with Prodigy InfoTech.

As part of the program, you agree to adhere to all company policies and guidelines applicable to non-employee interns. This letter outlines the complete understanding between you and Prodigy InfoTech regarding your internship and supersedes any prior discussions or agreements. Any modifications must be made in writing and signed by both parties.

We look forward to your participation in the internship program and wish you a rewarding and enriching experience with us.

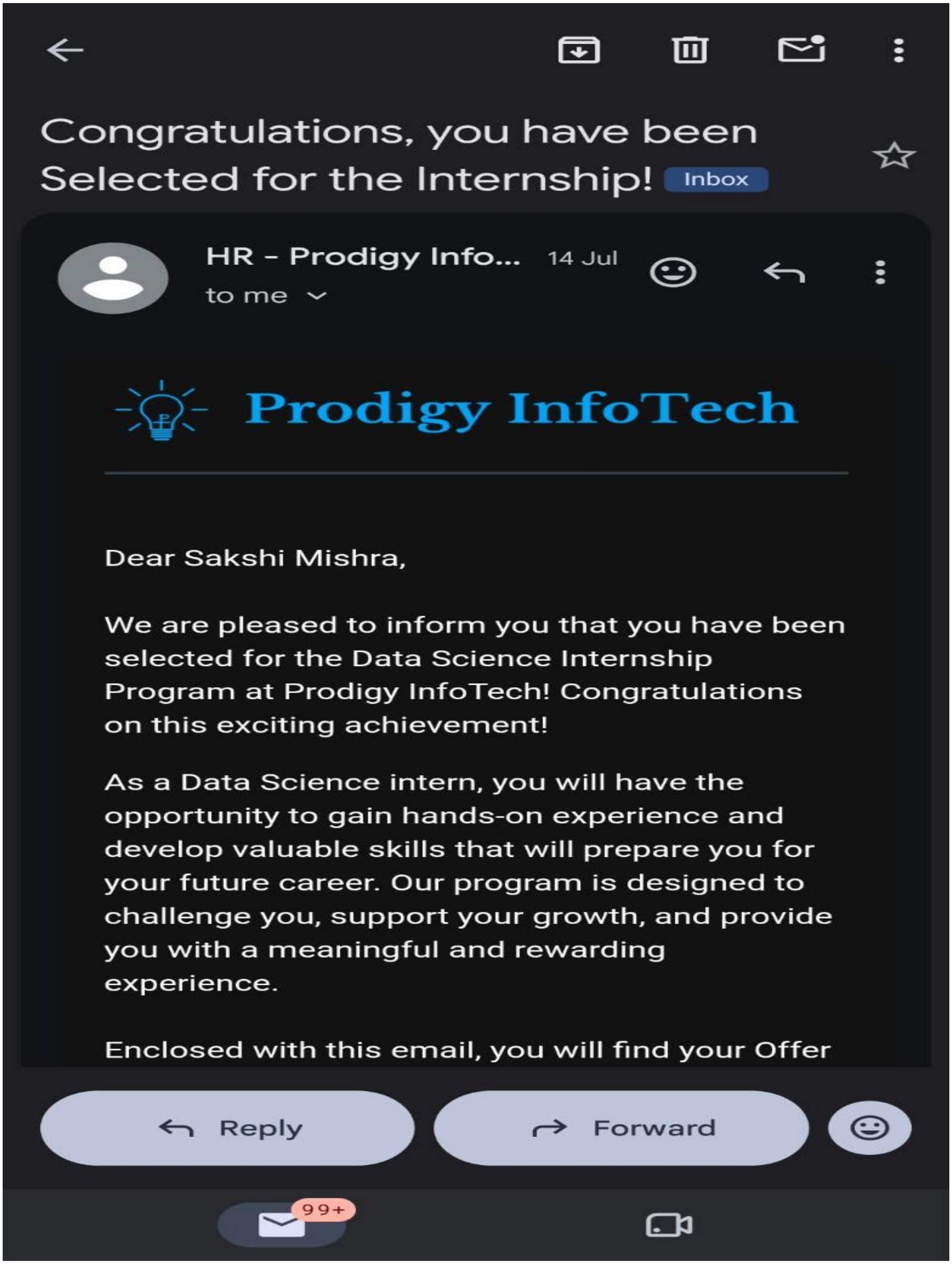
Sincerely,

Prodigy InfoTech

EMAIL
contact@prodigyinfotech.dev

WEBSITE.
prodigyinfotech.dev

Internship Offer/Selection Letter.



Internship Communication Proof.

Congratulations on the Successful Completion of Your Internship!

Inbox

HR - Prodigy Inf... 22 Aug

to me

Prodigy InfoTech

Dear Sakshi Mishra,

We extend our warmest congratulations on your successful completion of the Data Science internship program at Prodigy InfoTech. It is with great pleasure and pride that we acknowledge your hard work, dedication, and remarkable achievements throughout your internship journey.

As a token of appreciation for your outstanding performance, we are pleased to present you with a Certificate of Completion. This certificate not only recognizes your successful completion of

Reply

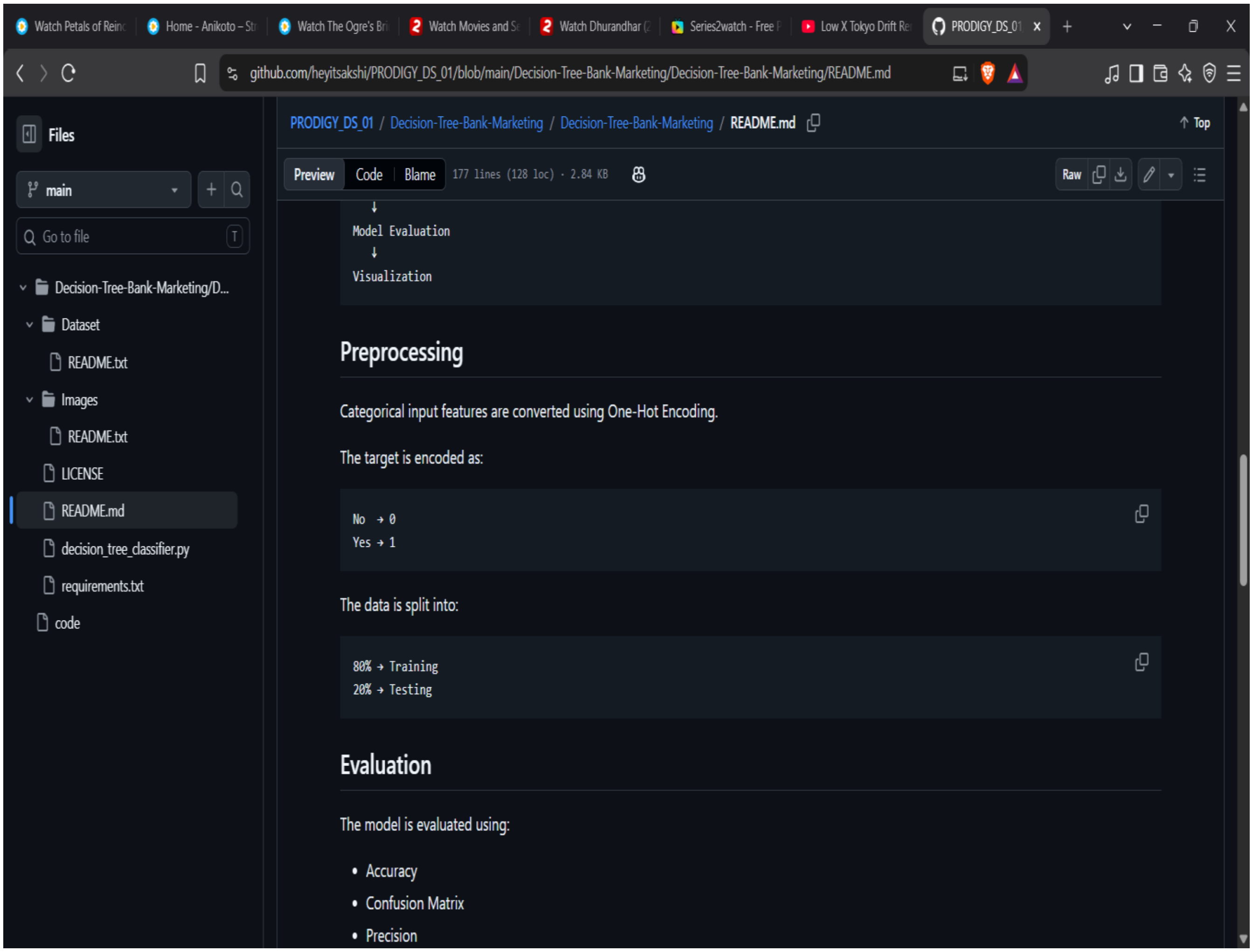
Forward

99+

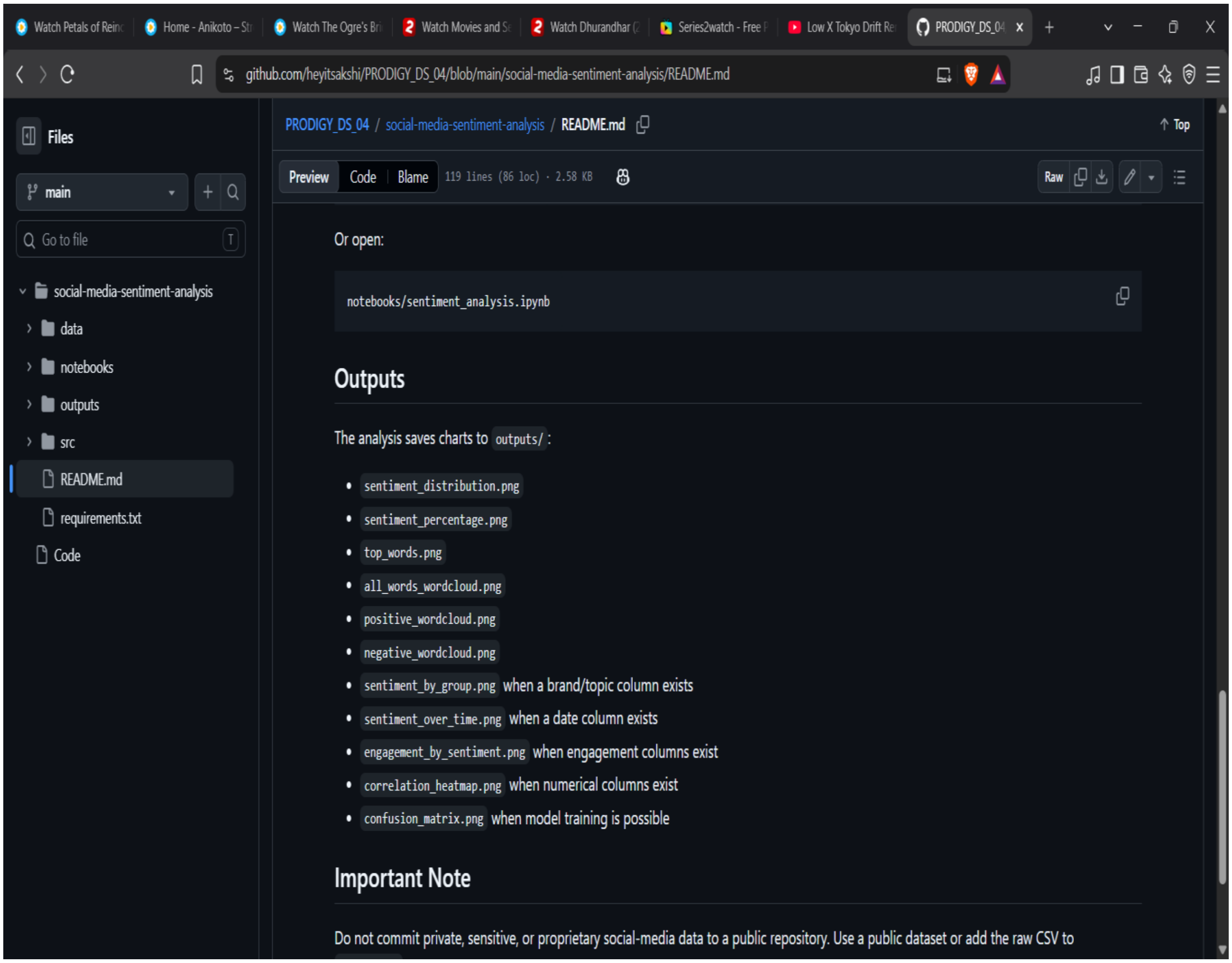
Project Completion Proof.

4.16 PROJECT SCREENSHOTS

Project 1 – Decision Tree for Bank Marketing

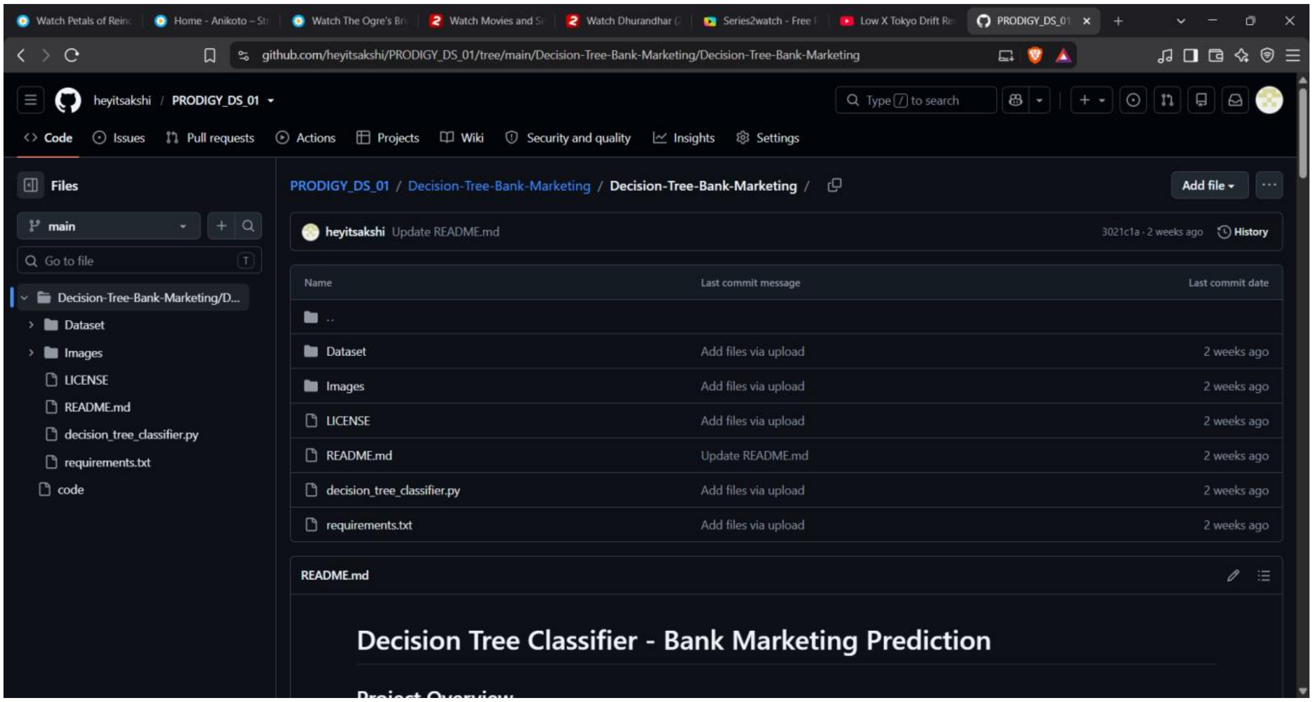


Project 2 – Social Media Statement Analysis

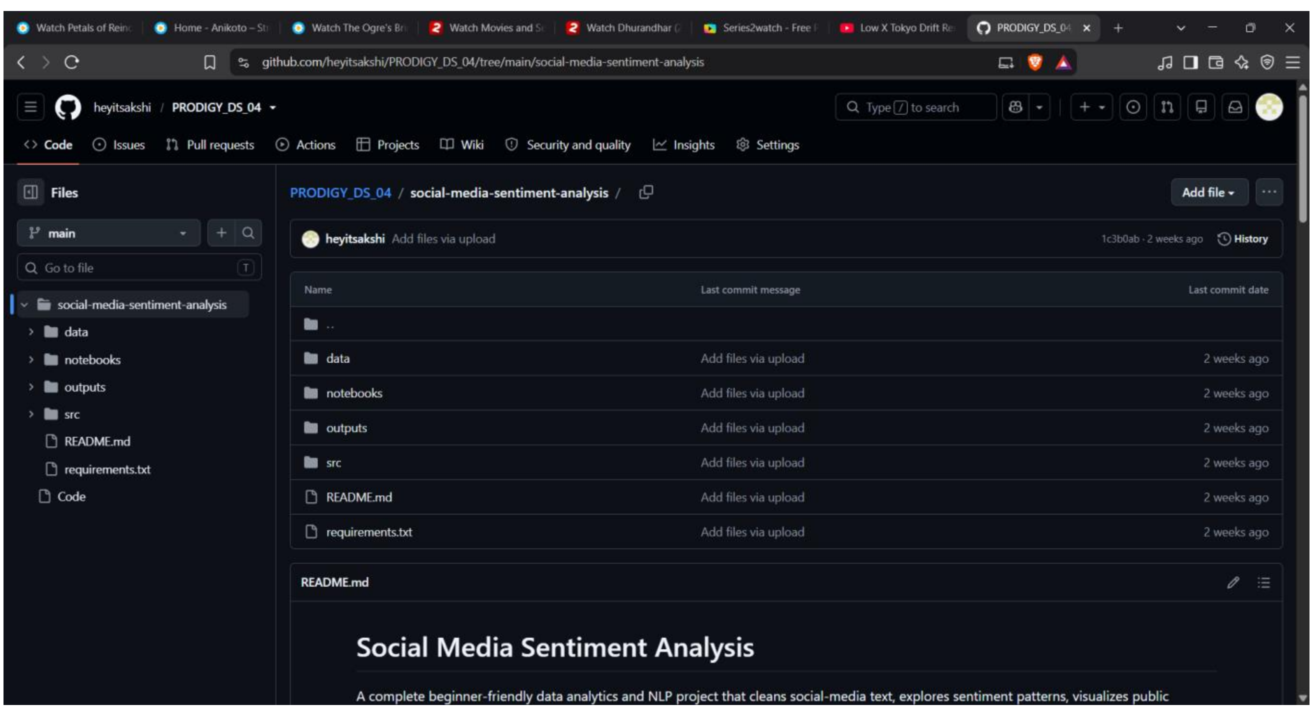


GITHUB REPOSITORY / PROJECT EVIDENCE

Decision Tree.



Social Media Sentiment Analysis.



5. BIBLIOGRAPHY / REFERENCES

1. Prodigy InfoTech, “Data Science Internship Program,” Internship Learning and Project Resources, 2026.

2. Python Software Foundation, “Python 3 Documentation,” Python Language and Standard Library Documentation.

3. Scikit-learn Documentation, “Decision Trees and Machine Learning Algorithms.”

4. Pandas Documentation, “Data Analysis and Manipulation Using Python.”

5. NumPy Documentation, “Numerical Computing with Python.”

6. Matplotlib Documentation, “Visualization with Python.”

7. Seaborn Documentation, “Statistical Data Visualization.”

8. UCI Machine Learning Repository, “Bank Marketing Dataset.”

9. Natural Language Processing and sentiment analysis learning resources.

10. GitHub Documentation, “Repository Management and Version Control.”